

BASIS

A set of vectors $\{v_1, v_2, \dots, v_k\}$ is said to be a basis for a subspace S if

(i) $\text{span}\{v_1, \dots, v_k\} = S$, AND

(ii) $\{v_1, \dots, v_k\}$ are linearly independent.

A subspace can have multiple bases:

Example:

$\mathcal{V} = \mathbb{F}^n$ Consider

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

Easy to verify that $\{e_1, e_2, \dots, e_n\}$ is a basis of $\mathcal{V}$.

An implication of linear dependence

If $v_1, \dots, v_m$ are a set of linearly dependent vectors, then

$$\sum_{i=1}^m \alpha_i v_i = 0 \text{ for some}$$

$\alpha_1, \dots, \alpha_m$, (not all of them zeros).

This means there is α_j (for some j between 1 and m) which is non-zero

$$\Rightarrow \alpha_j v_j = \mathbf{I}_I \sum_{\substack{i=1 \\ i \neq j}}^m \alpha_i v_i$$

$$\Rightarrow v_j = \mathbf{I}_I \alpha_j^{-1} \sum_{\substack{i=1 \\ i \neq j}}^m \alpha_i v_i \quad \left(\because \alpha_j \neq 0, \text{ its field inverse exists} \right)$$

ie v_j can be represented

as a linear combination of the other vectors

Basis

A set of vectors $\{v_1, \dots, v_k\}$ is said to be a basis for a subspace S if

$$S = \text{span}\{v_1, \dots, v_k\} \text{ AND}$$

$v_1, \dots, v_k$ are linearly independent.

Last lecture:

Basis for $\mathbb{F}^n$:

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$\{e_1, \dots, e_n\}$ is a basis for $\mathbb{F}^n$.

Existence of Basis

① Suppose a subspace S ($S \neq \{0\}$) has a finite spanning set, i.e.

$$S = \text{span}\{a_1, a_2, \dots, a_m\}$$

then S has a basis.

Proof: Without loss of generality, assume that

$a_1, a_2, \dots, a_m$ are all non-zero vectors.
(Otherwise, if any one of them is the zero vector, we can remove it from the set and the remaining vectors continue to be a spanning set of S).

Two cases arise: Either $\{a_1, a_2, \dots, a_m\}$ are linearly independent (LI) or they are dependent (LD). If $a_1, \dots, a_m$ are LI, we have our basis for S .
If $a_1, \dots, a_m$ are LD, then notice the following fact:

Fact: There exists j ($1 \leq j \leq m$)

s.t. $a_j = \sum_{\substack{i=1 \\ i \neq j}}^m \alpha_i a_i$, for some
 $\alpha_1, \dots, \alpha_{j-1}, \alpha_{j+1}, \dots, \alpha_m \in \mathbb{F}$

Why?

Since $a_1, \dots, a_m$ are LD, there exist scalars $\beta_1, \dots, \beta_m$, (not all zeros) such that

$$\sum_{i=1}^m \beta_i a_i = 0$$

$\Rightarrow$ There exists j ($1 \leq j \leq m$) such that $\beta_j \neq 0$.

$$\Rightarrow \beta_j a_j = 1 \cdot \sum_{\substack{i=1 \\ i \neq j}}^m \beta_i a_i$$

$$\Rightarrow a_j = \sum_{\substack{i=1 \\ i \neq j}}^m \underbrace{(1 \cdot \beta_i)}_{\triangleq \alpha_i} \beta_j^{-1} a_i$$

Hence, if we remove a_j from the set $\{a_1, \dots, a_m\}$, we obtain the set $I_1 \triangleq \{a_1, \dots, a_m\} \setminus \{a_j\}$

Convince yourself that I_1 continues to be a spanning set for S .

Obviously $|I_1| = m-1$.

We again have two cases: either I_1 is a basis of S , or we construct a set I_2 following previous argument where $\text{span}\{I_2\} = S$ and $|I_2| = m-2$

More generally, if the above procedure continues for " i " steps, then we have

$$|I_i| = m-i, \quad \text{span}\{I_i\} = S$$

The procedure terminates no later than $m-1$ steps. If it terminates at step $j < m-1$, I_j must be linearly independent and $\text{span}\{I_j\} = S$

$\Rightarrow I_j$ is a basis for S .

If the procedure continues upto step $m-1$, I_{m-1} will comprise of a single (non-zero) vector, which is linearly independent.

Moreover $\text{span}\{I_{m-1}\} = S$

$\Rightarrow I_{m-1}$ will be a basis.

② If $\{a_1, \dots, a_m\}$ is a linearly independent set in a subspace S and if

$\{b_1, \dots, b_p\}$ is a spanning set of S , then $m \leq p$. (Dimension-Independence Inequality)

Proof: Assume $m > p$.

Since $\{b_1, \dots, b_p\}$ is a spanning set of S , and $a_1 \in S$, there exist $\alpha_1, \dots, \alpha_p \in \mathbb{F}$ such that

$$a_1 = \sum_{i=1}^p \alpha_{1i} b_i$$

Note at least one of $\alpha_{11}, \dots, \alpha_{1p}$ must be non-zero
 (otherwise $a_1 = 0$, $\Rightarrow$ not possible since $a_1, \dots, a_m$ are
 Lin. indep.)

Suppose $\alpha_{1j} = 0$ for some j

$$\Rightarrow b_j = \sum_{\substack{i=1 \\ i \neq j}}^p \beta_i b_i + r_1 a_1 \quad \left(\begin{array}{l} \text{DIT:} \\ \text{figure out what} \\ \beta_i \text{ and } r_1 \text{ are} \end{array} \right)$$

$\Rightarrow$ We can replace b_j with a_1 in the spanning set and the resulting set will continue to be a spanning set.

[Denote $B = \{b_1, b_2, \dots, b_{j-1}, b_j, b_{j+1}, \dots, b_p\}$

$$\text{let } B_1 = B \setminus \{b_j\} \cup \{a_1\}$$

$$\Rightarrow \text{span}\{B_1\} = \text{span}\{B\} = S]$$

Now, $a_2 \in S$ and $\text{span}\{B_1\} = S$

$$\Rightarrow a_2 = \sum_{\substack{i=1 \\ i \neq j}}^p \alpha_{2i} b_i + \alpha'_1 a_1$$

Note, at least one of α_{2i} ($i=1, 2, \dots, p, i \neq j$) must

be non zero. Otherwise a_2 will be either linearly dependent with a_1 , or the zero vector.

let $\alpha_{2k} \neq 0$ for some k ($k \neq j$)

$\Rightarrow$ We can replace b_k with a_2 and obtain the set:

$$B_2 = B_1 \setminus \{b_k\} \cup \{a_2\}$$

with the property $\text{span}\{B_2\} = \text{span}\{B_1\} = S$

Proceeding this way, we can replace each of $b_1, \dots, b_p$ with $a_1, \dots, a_p$ (since we assumed $m > p$). Call this set B where

$$B_p = \{a_1, a_2, \dots, a_p\}, \text{span}\{B_p\} = S$$

Since $m > p$, there still remains a_{p+1} in the set of linearly independent vectors.

$$\Rightarrow a_{p+1} = \sum_{i=1}^p \alpha_i a_i \quad \left(\text{since } \{a_1, \dots, a_p\} \text{ spans } S \right)$$

$\Rightarrow$ CONTRADICTS that $\{a_1, \dots, a_p, a_{p+1}\}$ are LINEARLY INDEPENDENT

$\Rightarrow$ It is not possible to replace all elements in B with the linearly independent vectors and one

assumption $m > p$ is wrong

$$\Rightarrow m \leq p.$$

DIMENSION

(3) Suppose $B = \{b_1, \dots, b_{p_1}\}$ is a basis of S and $C = \{c_1, \dots, c_{p_2}\}$ is another basis of S .
then $p_1 = p_2$.

Proof:

B is LI set and C is spanning set

$$\Rightarrow |B| \leq |C|$$

C is LI and B is spanning set

$$|C| \leq |B|$$

$$\Rightarrow |C| = |B| \Rightarrow p_1 = p_2$$

The (common) cardinality of all bases of a subspace is called its "DIMENSION"

Examples;

① $V = \mathbb{F}^n$ $\dim(V) = n$.

② $V = \mathbb{F}^{m \times n}$

$$E_{11} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}, \quad E_{12} = \begin{bmatrix} 0 & 1 & 0 & \dots \\ 0 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix},$$

$$\dots, E_{ij} = \begin{bmatrix} 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 1 & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \text{ (with } 1 \text{ at } (i,j) \text{)}, \dots,$$

$$E_{mn} = \begin{bmatrix} 0 & \dots & 0 \\ \vdots & & \vdots \\ 0 & \dots & 1 \end{bmatrix}$$

$$\dim(\mathbb{F}^{m \times n}) = m \cdot n.$$

③ $\dim\{0\} = 0$ ($\{0\} = \text{span}\{0\}$. But 0 is not linearly indep.)

④ $V = \mathbb{R}^3$
 $S = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix}, x - 2y = 0 \right\}$

Consider $u \in S$ $u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 2u_2 \\ u_2 \\ u_3 \end{bmatrix} = u_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + u_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$S \subseteq \text{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Verify $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \in S$

$$\Rightarrow \text{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \subseteq S \quad (\because S \text{ is a subspace})$$

$$S = \text{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Verify $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are lin. indep.

$$\boxed{\dim(S) = 2}$$

Recall:

$$f: \mathbb{F}^n \rightarrow \mathbb{F}^m$$

$$A \in \mathbb{F}^{m \times n} \text{ (given)}$$

$$f(x) = Ax$$

$$R(A) = \left\{ y \in \mathbb{F}^m, \text{ s.t. } y = Ax \text{ for some } x \in \mathbb{F}^n \right\}$$

$R(A)$ is a subspace

$$\dim(R(A)) = \text{rank}(A)$$

(Definition of Rank)

Linear Map

Let U, V be vector spaces defined over the same field $\mathbb{F}$.

A map $f: U \rightarrow V$ is said to be linear

if $f(u+v) = f(u) + f(v)$, $u, v \in U$

addition over U *Addition over V*

and $f(\alpha \cdot v) = \alpha \cdot f(v)$, $\alpha \in \mathbb{F}, v \in U$

scalar mult. of U *scalar mult. of V*